

FORT SMITH, NT, Canada

WMO: 719340

Lat: 60.026N

Lon: 111.929W

Elev: 673

StdP: 14.34

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.4	-31.8	-40.8	0.5	-33.6	-36.8	0.7	-30.1	16.5	1.3	14.6	0.0	1.8	300	0.661

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	82.7	62.7	79.7	61.2	76.7	59.7	65.2	77.9	63.4	75.2	61.8	73.0	7.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.8	81.7	68.5	58.8	76.0	66.9	57.1	71.3	65.7	30.5	78.2	29.2	75.6	28.0	73.4	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.2	14.1	12.5	-41.1	88.4	3.7	2.7	-43.8	90.3	-46.0	91.9	-48.1	93.4	-50.8	95.3		
			-39.2	69.2	2.8	3.0	-41.2	71.4	-42.9	73.1	-44.4	74.8	-46.5	77.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.7	-7.3	-1.7	10.7	30.6	46.8	59.1	63.3	59.1	48.3	33.4	13.6	-1.2
	DBStd	27.39	15.16	13.61	15.84	12.65	10.09	7.33	5.80	6.65	8.02	8.43	12.05	15.04
	HDD50	8549	1777	1448	1219	586	177	13	0	8	125	516	1093	1587
	HDD65	12974	2242	1868	1684	1032	568	200	102	204	501	978	1543	2052
	CDD50	1147	0	0	0	4	77	287	413	289	74	3	0	0
	CDD65	98	0	0	0	0	3	24	50	20	1	0	0	0
	CDH74	1247	0	0	0	0	74	338	554	260	21	0	0	0
	CDH80	233	0	0	0	0	8	56	112	55	2	0	0	0
Wind	WSAvg	5.3	4.4	4.8	5.4	5.9	6.1	5.4	5.0	5.2	5.7	5.5	5.1	4.7
Precipitation	PrecAvg	14.4	0.8	0.6	0.6	0.6	1.2	1.8	2.3	2.1	1.5	1.2	0.9	0.8
	PrecMax	21.9	1.7	1.3	1.4	1.8	2.8	5.5	6.1	5.7	3.1	4.3	2.0	1.7
	PrecMin	9.2	0.0	0.0	0.0	0.0	0.1	0.4	0.3	0.3	0.0	0.2	0.1	0.2
	PrecStd	2.7	0.4	0.3	0.3	0.4	0.7	1.3	1.1	1.1	0.7	0.7	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.1	35.1	50.4	65.8	80.6	85.2	86.9	86.3	77.2	61.8	43.3	32.5
		MCWB	28.4	31.9	39.4	48.6	57.7	62.7	65.2	65.6	60.6	50.6	36.8	30.4
	2%	DB	24.4	27.9	43.7	59.3	75.7	81.2	83.0	80.3	70.8	54.2	34.9	27.0
		MCWB	23.4	26.2	35.7	44.5	54.5	60.6	63.5	62.2	57.4	45.8	32.8	25.9
	5%	DB	18.4	22.3	38.5	54.2	69.9	77.9	80.1	76.4	65.6	49.2	32.5	22.8
		MCWB	17.5	20.8	32.7	42.4	52.1	59.0	62.5	60.6	54.8	42.8	31.4	21.9
	10%	DB	13.2	17.8	33.4	49.4	64.6	74.6	76.8	72.6	61.7	44.8	29.3	19.0
		MCWB	12.5	16.5	29.7	39.6	49.6	58.0	61.3	59.4	52.7	40.6	28.3	18.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.6	32.4	39.6	49.6	59.2	65.7	68.2	67.3	61.6	52.5	37.1	30.8
		MCDB	29.6	35.4	49.2	63.5	78.7	79.4	79.2	81.8	74.1	60.2	42.3	32.2
	2%	WB	23.7	26.1	36.0	45.6	55.8	63.3	65.8	64.2	58.5	46.3	33.2	25.9
		MCDB	24.7	27.8	43.0	57.4	71.8	76.1	78.8	75.7	68.2	53.0	34.7	27.1
	5%	WB	17.8	20.9	32.8	42.7	53.2	61.2	64.1	62.3	56.1	43.4	31.4	22.0
		MCDB	18.5	22.3	38.0	53.6	67.6	73.5	75.8	73.0	63.9	48.5	32.7	22.8
	10%	WB	12.7	16.6	29.7	39.9	50.5	59.4	62.6	60.6	53.8	40.7	28.3	18.3
		MCDB	13.4	17.8	33.5	48.9	63.5	71.1	74.0	70.1	60.6	44.5	29.2	19.1
Mean Daily Temperature Range	5% DB	MDBR	14.1	18.1	21.9	21.3	22.1	21.9	20.6	19.8	17.7	12.2	11.2	13.1
		MCDBR	17.1	21.2	23.6	26.3	29.5	27.2	25.9	26.7	25.0	20.0	11.3	14.6
		MCWBR	16.6	19.5	16.9	15.2	14.5	12.3	11.7	12.4	13.9	12.8	9.4	13.8
	5% WB	MCDBR	17.3	20.8	23.0	25.0	26.5	22.6	22.5	22.9	22.0	18.4	11.0	14.2
MCWBR		16.8	19.2	16.9	14.8	13.6	11.1	11.3	11.9	12.8	12.1	9.3	13.6	
Clear-Sky Solar Irradiance	taub	0.211	0.245	0.267	0.317	0.338	0.360	0.393	0.355	0.314	0.278	0.218	0.171	
	taud	2.143	2.276	2.284	2.311	2.434	2.389	2.350	2.445	2.569	2.548	2.240	2.052	
	Ebn at Noon	191	250	278	280	282	276	263	266	262	239	188	164	
	Edh at Noon	13	20	28	33	32	34	35	29	22	16	11	9	
All-Sky Solar Radiation	RadAvg	141	435	981	1566	1827	1938	1735	1369	872	395	132	74	
	RadStd	12	23	47	94	169	156	104	69	75	39	20	9	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page